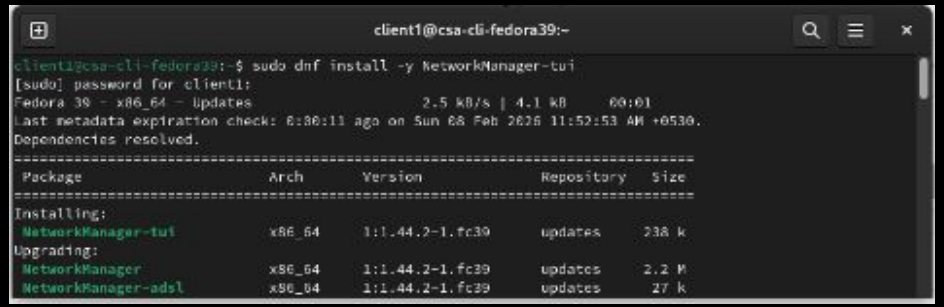
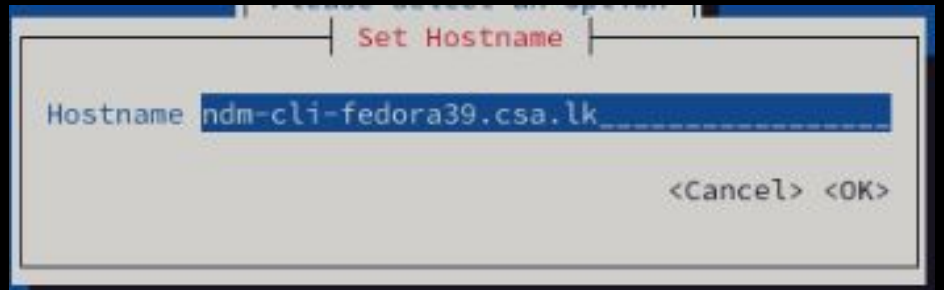
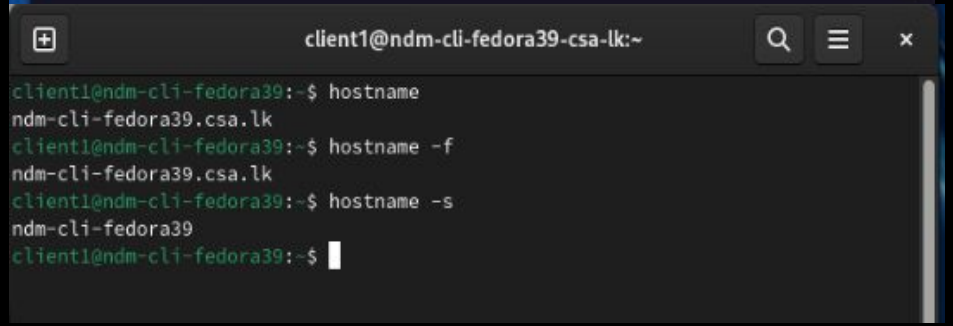
1. vi <file> → Open a file in vi editor
2. Press i → Switch to insert mode (edit the file)
3. Press Esc → Exit insert mode
4. :w → Save changes
5. :q → Quit the editor
6. :wq or ZZ → Save and quit
7. :q! → Quit without saving
8. cd <directory> → Change directory
9. ls → List files and directories
10. cd -To Go Back to the Previous Directory
11. sudo <command> → Run a command as root
12. su - <user> → Switch to another user

Install (NetworkManager-tui) in client (fedora) so we can manage network settings using the nmtui command.







Run- sudo dnf install -y NetworkManager-tui

After installation, run:- nmtui

Select “Set system hostname”

Enter- ndm-cli-fedora39.csa.lk

Press OK

Exit nmtui

Then verify:

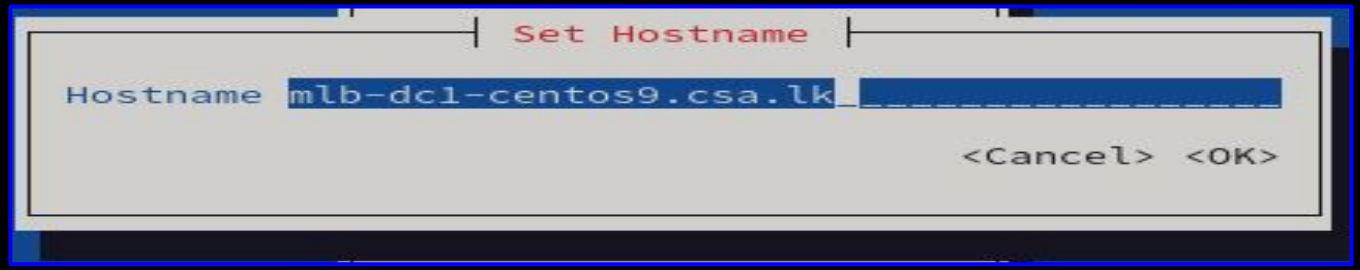
hostname
hostname -f

Change hostname in server (centOS)

Stop dhcp service - service dhcpd stop

Change hostname in to - mlb-dc1-centos9.csa.lk

nmtui > set system hostname >



verify the change in host name - hostname
Fully Qualified Domain Name (FQDN) - hostname -f
Short Hostname - hostname -s

```
[root@mlb-dc1-centos7 ~]# hostname
mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos7 ~]# hostname -f
mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos7 ~]# hostname -s
mlb-dc1-centos9
```

Edit the hosts file as follows vi /etc/hosts

```
root@mlb-dc1-centos7:~ — /usr/bin/vim /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
10.0.1.2    mlb-dc1-cenntos9.csa.lk
```

Reboot your machine to make this changes in to an effect

install DNS server on CentOS 9, enter the following command:

nmtui>activate >NAT

dnf install -y bind bind-utils

Start the service up - service named start

Check the status of the named service - service named status

Stop your name server service and dhcp service if they are running

```
[root@server ~]#service dhcpd stop
 [root@server ~]#service named stop
```

Open the BIND configuration file (/etc/named.conf), add a forward and reverse zone for your network, then save and close the file.

Open the configuration files of the dns server - vi /etc/named.conf

```
options {
        listen-on port 53 { 127.0.0.1; 10.0.1.2; };
        listen-on-v6 port 53 { ::1; };
        directory        "/var/named";
        dump-file        "/var/named/data/cache_dump.db";
        statistics-file "/var/named/data/named_stats.txt";
        memstatistics-file "/var/named/data/named_mem_stats.txt";
        recursing-file  "/var/named/data/named.recursing";
        secroots-file   "/var/named/data/named.secroots";
        allow-query     { localhost; 10.0.1.0/24; };
```

Step 1

vi /etc/named.conf

```
  GNU nano 2.3.1                                  File:

};

zone "." IN {
        type hint;
        file "named.ca";
};

## Forward lookup zone

zone "csa.lk" IN {
     type master;
     file "forward.csa.lk";
     allow-update { none; };
};



## Reverse lookup zone

zone "1.0.10.in-addr.arpa" IN {
     type master;
     file "reverse.csa.lk";
     allow-update { none; };
};
_

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
```

```
## forward lookup zone
zone "csa.lk" IN {
  type master;
  file "forward.csa.lk";
  allow-update { none; };
};
```

```
## Reverse lookup zone
zone "1.0.10.in-addr.arpa" IN {
  type master;
  file "reverse.csa.lk";
  allow-update { none; };
};
```

Step 3

Open a new file named forward.csa.lk in the named directory as follows to create the forward zone file for the DNS server. Replace “ ? “ with correct values

vi /var/named/forward.csa.lk

root@mlb-dc1-centos9:~ — /usr/bin/vim /var/name...

```
$TTL 86400
@   IN  SOA  mlb-dc1-centos9.csa.lk. root.csa.lk. (
        2011071001  ; Serial
        3600        ; Refresh
        1800        ; Retry
        604800      ; Expire
        86400       ; Minimum TTL
)
@   IN  NS   mlb-dc1-centos9.csa.lk.
@   IN  A    10.0.1.2
@   IN  A    10.0.1.3

mlb-dc1-centos9    IN  A   10.0.1.2
ndm-cli-fedora39   IN  A   10.0.1.3
```

```
$TTL 86400
@   IN  SOA  mlb-dc1-centos9.csa.lk. root.csa.lk. (
        2011071001  ; Serial
        3600        ; Refresh
        1800        ; Retry
        604800      ; Expire
        86400       ; Minimum TTL
)
@   IN  NS   mlb-dc1-centos9.csa.lk.
@   IN  A    10.0.1.?
@   IN  A    10.0.1.?

mlb-dc1-centos9    IN  A   10.0.1.?
?   IN  A   10.0.1.?
```

Open a new file named reverse.csa.lk in the named directory as follows to create the reverse zone file for the DNS server.

vi /var/named/reverse.csa.lk

```
root@mlb-dc1-centos9:~ — /usr/bin/vim /var/name..

$TTL 86400
@ IN SOA mlb-dc1-centos9.csa.lk. root.csa.lk. (
    2011071001  ; Serial
    3600            ; Refresh
    1800            ; Retry
    604800         ; Expire
    86400           ; Minimum TTL
)

@   IN NS mlb-dc1-centos9.csa.lk.
@   IN PTR csa.lk.
mlb-dc1-centos9 IN A 10.0.1.2
ndm-cli-fedora39 IN A 10.0.1.3
2 IN PTR mlb-dc1-centos9.csa.lk.
3 IN PTR ndm-cli-fedora39.csa.lk.
```

```
$TTL 86400
@ IN SOA mlb-dc1-centos9.csa.lk. root.csa.lk. (
    2011071001  ; Serial
    3600            ; Refresh
    1800            ; Retry
    604800         ; Expire
    86400           ; Minimum TTL
)

@   IN NS mlb-dc1-centos9.csa.lk.
@   IN PTR csa.lk.
? IN A 10.0.1.?
? IN A 10.0.1.?
? IN PTR mlb-dc1-centos9.csa.lk.
? IN PTR ?
```

Open the DHCP server configuration file (/etc/dhcp/dhcpd.conf), set the domain name and DNS server, define the subnet and IP range for your network, then save and close the file.

vi /etc/dhcp/dhcpd.conf

root@mlb-dc1-centos9:~ — /usr/bin/vim /etc/dhcp/...

```
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#

# option definitions common to all supported networks...
option domain-name "csa.lk";
option domain-name-servers mlb-dc1-centos9.csa.lk;

default-lease-time 600;
max-lease-time 7200;

# Use this to enble / disable dynamic dns updates globally.
#ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

# Use this to send dhcp log messages to a different log file (you also
# have to hack syslog.conf to complete the redirection).
log-facility local7;

# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.

subnet 10.152.187.0 netmask 255.255.255.0 {
}
-- INSERT --                                    1,1           Top
```

root@mlb-dc1-centos9:~ — /usr/bin/vim /etc/dhcp/...

```
# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.

subnet 10.254.239.32 netmask 255.255.255.224 {
  range dynamic-bootp 10.254.239.40 10.254.239.60;
  option broadcast-address 10.254.239.31;
  option routers rtr-239-32-1.example.org;
}

# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
  range 10.0.1.25 10.0.1.50;
  option domain-name-servers mlb-dc1-centos9.csa.lk;
  option domain-name "csa.lk";
  option routers 10.0.1.1;
  option broadcast-address 10.0.1.255;
  default-lease-time 600;
  max-lease-time 7200;
}

# Hosts which require special configuration options can be listed in
# host statements.   If no address is specified, the address will be
# allocated dynamically (if possible), but the host-specific informati
on
# will still come from the host declaration.

host passacaglia {
  hardware ethernet 0:0:c0:5d:bd:95;
-- INSERT --                                    42,1          46%
```

allow DNS traffic through CentOS firewall by opening UDP port 53 permanently and then reload the firewall rules.

```
firewall-cmd --permanent --add-port=53/udp
```

```
firewall-cmd --reload
```

root@mlb-dc1-centos9:~

```
[root@mlb-dc1-centos9 ~]# firewall-cmd --permanent --add-port=53/udp
Warning: ALREADY_ENABLED: 53:udp
success
[root@mlb-dc1-centos9 ~]# firewall-cmd --reload
success
```

Check the DNS configuration and zone files for syntax errors using named-checkconf for the main config and named-checkzone for forward and reverse zones, ensuring all return “OK” if there are no errors.

named-checkconf /etc/named.conf

named-checkzone csa.lk /var/named/forward.csa.lk

named-checkzone csa.lk /var/named/reverse.csa.lk

root@mlb-dc1-centos9:~

```
[root@mlb-dc1-centos9 ~]# named-checkconf /etc/named.conf
[root@mlb-dc1-centos9 ~]# named-checkzone csa.lk /var/named/forward.csa.lk
zone csa.lk/IN: loaded serial 2011071001
OK
[root@mlb-dc1-centos9 ~]# named-checkzone csa.lk /var/named/reverse.csa.lk
zone csa.lk/IN: loaded serial 2011071001
OK
[root@mlb-dc1-centos9 ~]#
```

Open the NetworkManager configuration file of the vmnet2 network interface to manually edit its network settings.

```
root@mlb-dc1-centos7:/etc/NetworkManager/system-connections
[root@mlb-dc1-centos7 ~]# cd /etc/NetworkManager/system-connections/
[root@mlb-dc1-centos7 system-connections]# ls
ens33.nmconnection  vmnet2.nmconnection
[root@mlb-dc1-centos7 system-connections]# vi vmnet2.nmconnection
```

cd /etc/NetworkManager/system-connections/

➡ Show available network profiles

ls

➡ Open the vmnet2 network configuration file to edit:

vi ?

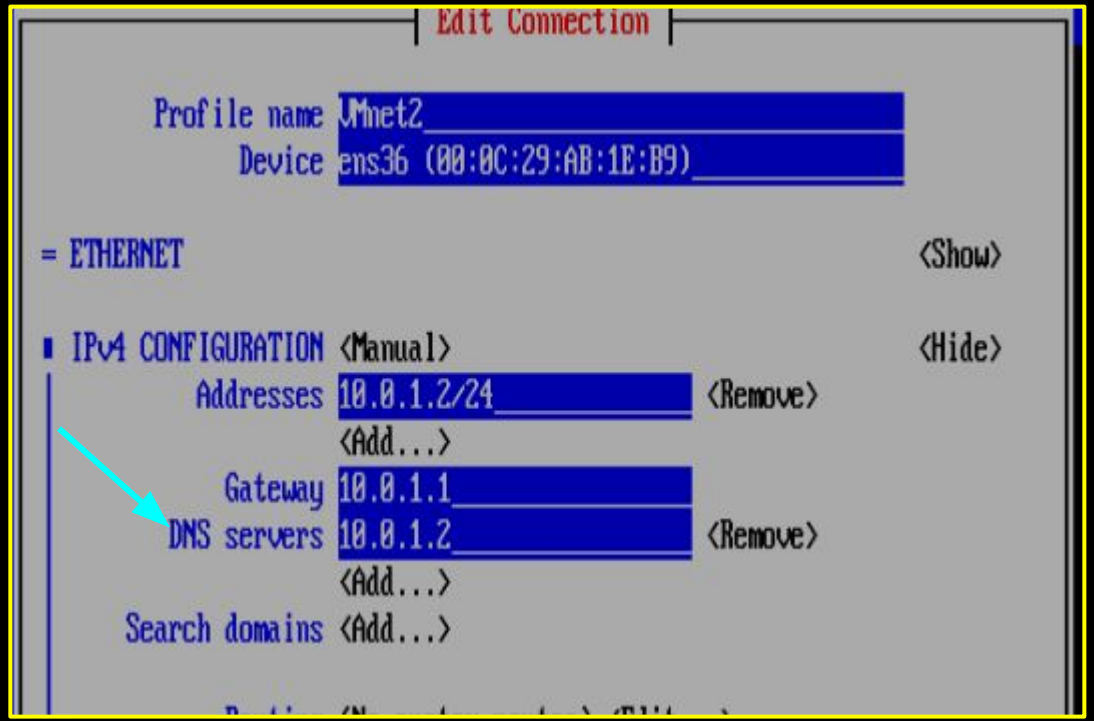
```
root@mlb-dc1-centos7:/etc/NetworkManager/system-connect...
  GNU nano 5.6.1                vmnet2.nmconnection                Modified
[connection]
id=vmnet2
uuid=3b20a48e-4cdf-4dce-9fed-ce76eca8eb8b
type=ethernet
interface-name=ens36
timestamp=1770791774

[ethernet]

[ipv4]
address1=10.0.1.2/24
gateway=10.0.1.1
dns=10.0.1.2
method=manual

[ipv6]
addr-gen-mode=default
method=auto

[proxy]

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Add dns address under ipv4

dns=10.0.1.2

Step 1

can do the same with nmtui >edit >vmnet2> add dns server > 10.0.1.2



These steps For centos7 only

```
Edit Connection

Profile name vmnet2
     Device ens36 (00:0C:29:AB:1E:B9)

= ETHERNET                                          <Show>

■ IPv4 CONFIGURATION <Manual>                       <Hide>
        Addresses 10.0.1.2/24          <Remove>
                  <Add...>
          Gateway 10.0.1.1
```

go to nmtui > edit > get a pictures of vmNet2 device no

```
[root@mlb-dc1-centos7 ~]# cd /etc/sysconfig/network-scripts/
[root@mlb-dc1-centos7 network-scripts]# ls
ifcfg-ens33  ifdown-ippp    ifdown-sit      ifup-bnep  ifup-plusb   ifup-TeamPort
ifcfg-ens36  ifdown-ipv6    ifdown-Team     ifup-eth   ifup-post    ifup-tunnel
ifcfg-lo     ifdown-isdn    ifdown-TeamPort ifup-ippp  ifup-ppp     ifup-wireless
ifdown       ifdown-post    ifdown-tunnel   ifup-ipv6  ifup-routes  init.ipv6-global
ifdown-bnep  ifdown-ppp     ifup            ifup-isdn  ifup-sit     network-functions
ifdown-eth   ifdown-routes  ifup-aliases    ifup-plip  ifup-Team    network-functions-ipv6
[root@mlb-dc1-centos7 network-scripts]# vi ifcfg-ens36
```

cd /etc/sysconfig/network-scripts/
cd /etc/NetworkManager/system-connections/ (for centos9)
ls
vi ifcfg-?

Replace “?” with vmnet2 deviceNo

```
IPADDR=10.0.1.2
PREFIX=24
GATEWAY=10.0.1.1
DNS=10.0.1.2
```

add to bottom line DNS=10.0.1.2

Step 1

Edit /etc/resolv.conf to set the local domain search path and specify your DNS server's IP so the system knows where to resolve hostnames.

vi /etc/resolv.conf

Step 2

```
root@mlb-dc1-centos9:~ — /usr/bin/vim /etc/resolv...
# Generated by NetworkManager
search mlb-dc1-centos9.csa.lk
nameserver 10.0.1.2
~
```

search mlb-dc1-centos9.csa.lk
nameserver 10.0.1.2

activate NAT & run

yum install bind-utils

Step 3

Restart your dhcp service and dns services.

nmtui>activate >vmnet2

nmtui>deactivate >nat

```
[root@server ~]#service dhcpd start
[root@server ~]]#service named start
```

Or

```
[root@server ~]#service dhcpd restart
[root@server ~]#service named restart
```

Note: If there are no errors in the zone files and the dhcp and dns configuration files both the operations should run smoothly.

Test DNS Server by using the following commands and check whether the input gets resolved correctly.

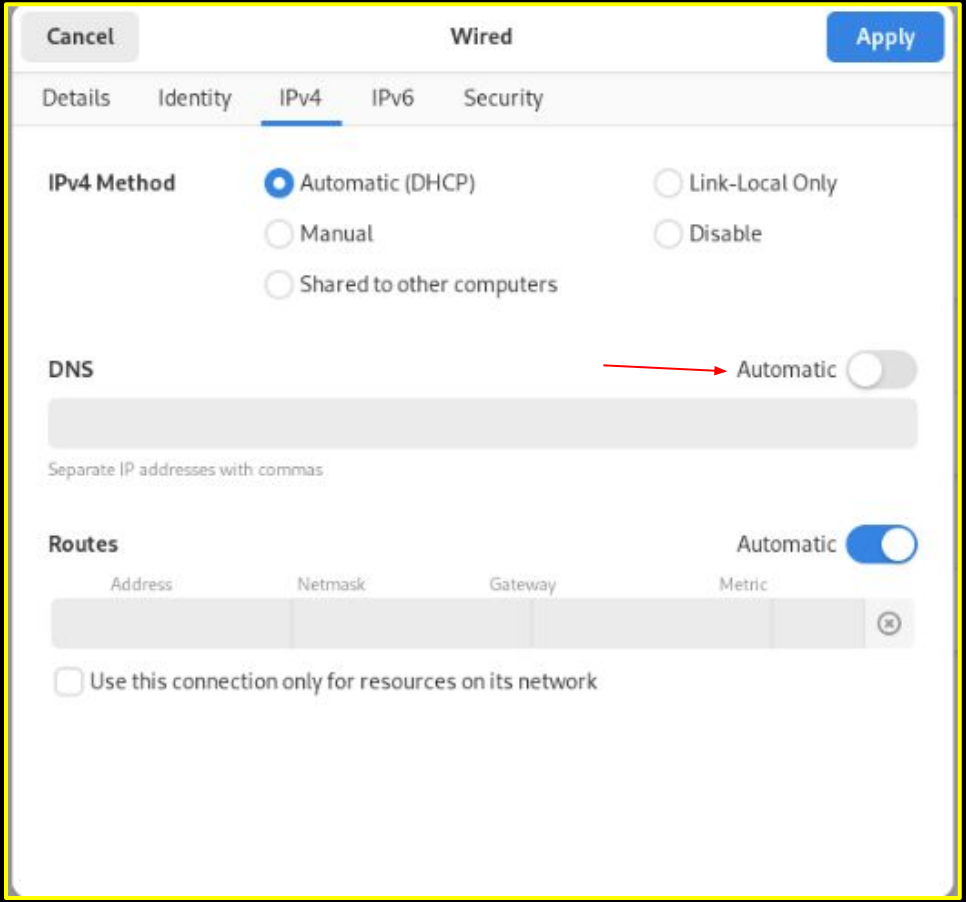
dig csa.lk

```
[root@mlb-dc1-centos9 ~]# dig csa.lk

; <<>> DiG 9.16.23-RH <<>> csa.lk
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 14599
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL:
1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 721b8e09f87b27e801000000698d4ef2d125d89da805b792 (good)
;; QUESTION SECTION:
;csa.lk.                        IN      A

;; ANSWER SECTION:
csa.lk.                 86400   IN      A       10.0.1.2
csa.lk.                 86400   IN      A       10.0.1.3

;; Query time: 1 msec
;; SERVER: 10.0.1.2#53(10.0.1.2)
;; WHEN: Thu Feb 12 09:24:26 +0530 2026
;; MSG SIZE  rcvd: 95

[root@mlb-dc1-centos9 ~]#
```

dig mlb-dc1-centos9.csa.lk

```
[root@mlb-dc1-centos9 ~]# dig mlb-dc1-centos9.csa.lk

; <<>> DiG 9.16.23-RH <<>> mlb-dc1-centos9.csa.lk
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 31836
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL:
1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: a70741e31df5375e01000000698d4f4c1c945de9eebcb31c (good)
;; QUESTION SECTION:
;mlb-dc1-centos9.csa.lk.                IN      A

;; ANSWER SECTION:
mlb-dc1-centos9.csa.lk. 86400   IN      A       10.0.1.2

;; Query time: 0 msec
;; SERVER: 10.0.1.2#53(10.0.1.2)
;; WHEN: Thu Feb 12 09:25:56 +0530 2026
;; MSG SIZE  rcvd: 95

[root@mlb-dc1-centos9 ~]#
```

Go to the (Fedora) client (vmNet2) settings



Add DNS server details on the (fedora)client side resolv.conf file

sudo vi /etc/resolv.conf

client1@fedora:~ — sudo nano /etc/resolv.conf

```
GNU nano 7.2                    /etc/resolv.conf                    Modified
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 10.0.1.2
options edns0 trust-ad
search mlb-dc1-centos9.csa.lk
```

search mlb-dc1-centos9.csa.lk
nameserver 10.0.1.2

Again, test DNS Server from the (fedora) client side by using the following commands and check whether the input gets resolved correctly.

dig csa.lk

client1@ndm-cli-fedora28:~

```
ndm-cli-fedora28.ndm.lk
[client1@ndm-cli-fedora28 ~]$ dig csa.lk

; <<>> DiG 9.18.7 <<>> csa.lk
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 27806
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 1, ADDITIONAL: 2

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 6c1f002e2278314b00fbf76367c88504a5020c0acaa65d3c (good)
;; QUESTION SECTION:
;csa.lk.                        IN      A

;; ANSWER SECTION:
csa.lk.                 86400   IN      A       10.0.1.3
csa.lk.                 86400   IN      A       10.0.1.2

;; AUTHORITY SECTION:
csa.lk.                 86400   IN      NS      mlb-dc1-centos7.csa.lk.

;; ADDITIONAL SECTION:
mlb-dc1-centos7.csa.lk. 86400   IN      A       10.0.1.2

;; Query time: 2 msec
;; SERVER: 10.0.1.2#53(10.0.1.2) (UDP)
;; WHEN: Wed Mar 05 22:38:17 +0530 2025
```

dig mlb-dc1-centos9.csa.lk

client1@ndm-cli-fedora28:~

```
[client1@ndm-cli-fedora28 ~]$ dig mlb-dc1-centos7.csa.lk

; <<>> DiG 9.18.7 <<>> mlb-dc1-centos7.csa.lk
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 60427
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 7b44a91d8da8137b83b47bb767c885385c399800ed777f04 (good)
;; QUESTION SECTION:
;mlb-dc1-centos7.csa.lk.                IN      A

;; ANSWER SECTION:
mlb-dc1-centos7.csa.lk. 86400   IN      A       10.0.1.2

;; AUTHORITY SECTION:
csa.lk.                 86400   IN      NS      mlb-dc1-centos7.csa.lk.

;; Query time: 2 msec
;; SERVER: 10.0.1.2#53(10.0.1.2) (UDP)
;; WHEN: Wed Mar 05 22:39:10 +0530 2025
;; MSG SIZE  rcvd: 109

[client1@ndm-cli-fedora28 ~]$ dig mlb-dc1-centos7.csa.lk

; <<>> DiG 9.18.7 <<>> mlb-dc1-centos7.csa.lk
```